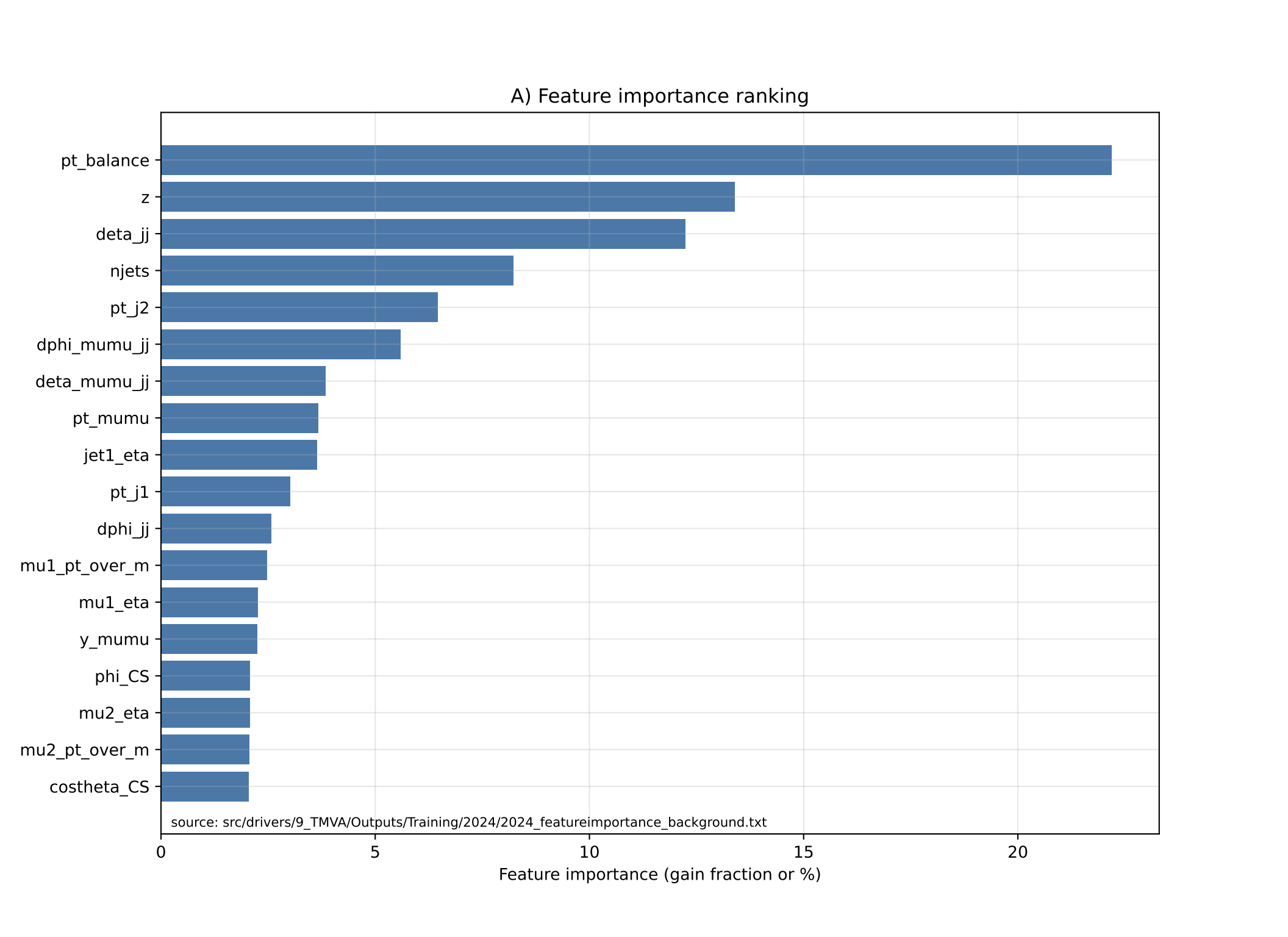
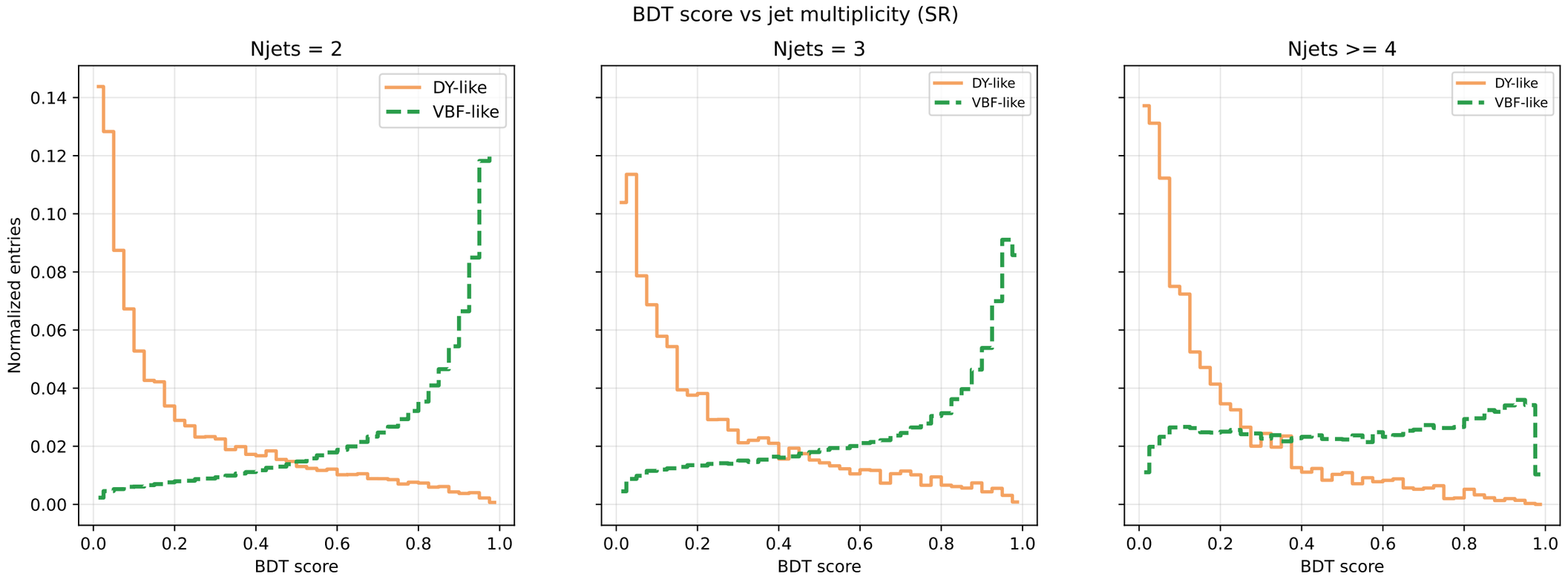
VBF BDT Validation - 2024

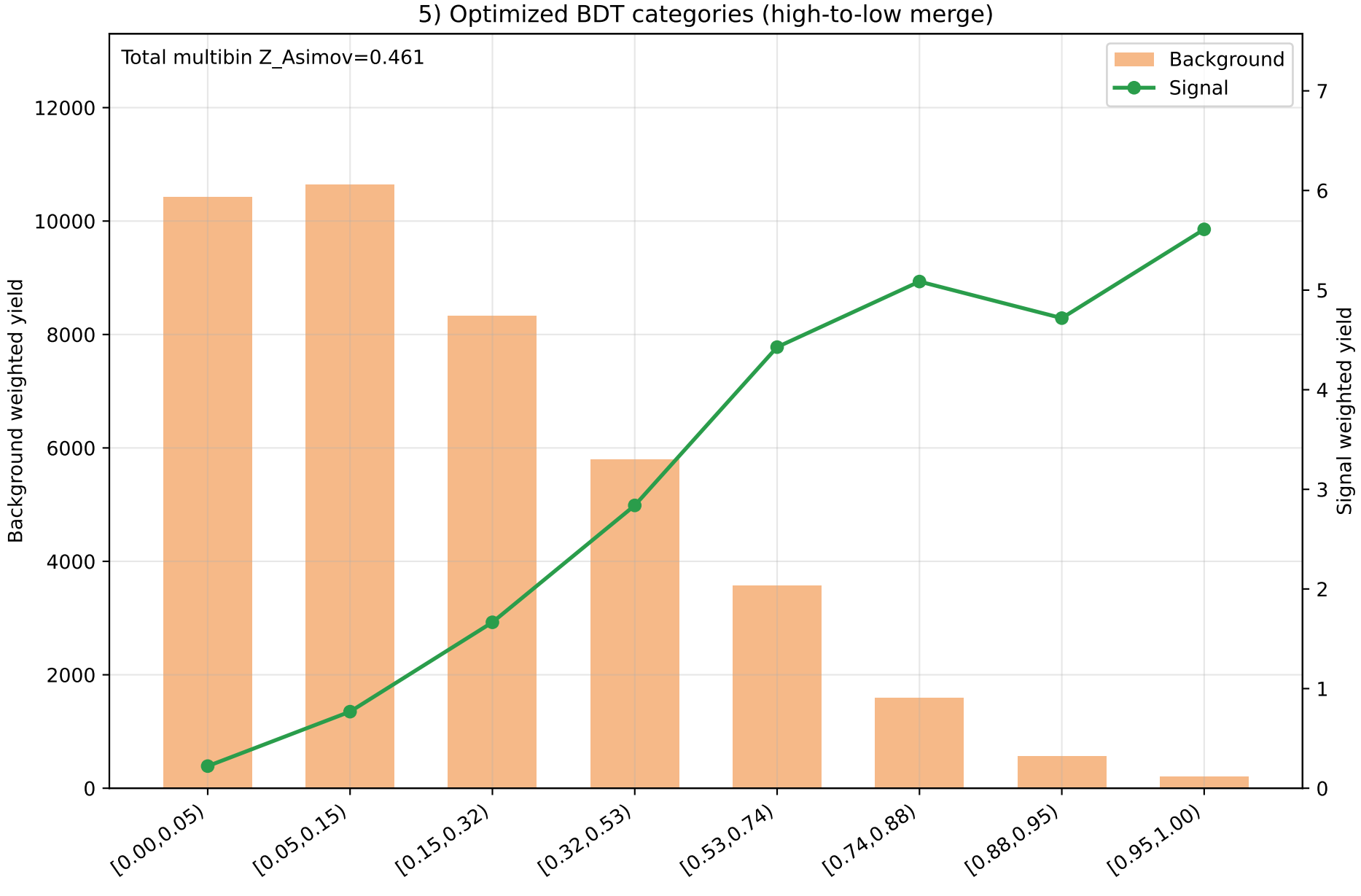
```
Standalone per-era diagnostic script: bdt_validation.py
Logic references: ScanYields / PlotScored / MassSculpt / KS_perfold / DataMC
Resolved branches: {'score': 'bdt_score', 'weight': 'weight', 'mass': 'm_mumu', 'njets': 'njets', 'mjj': None, 'de
```

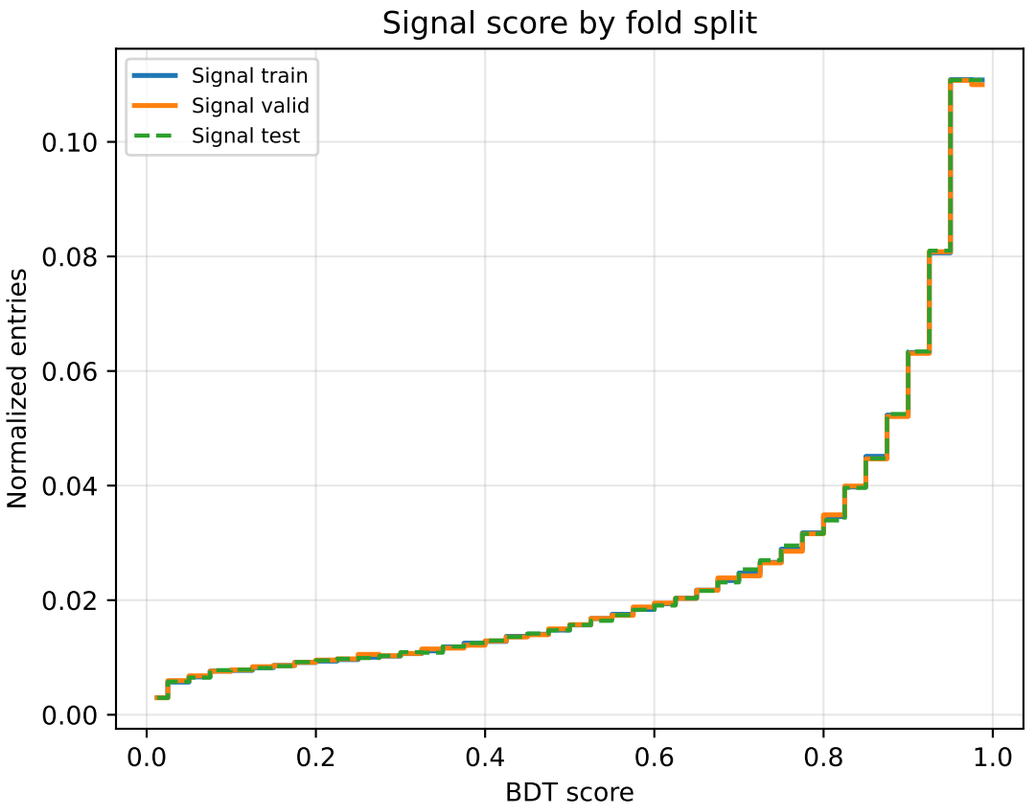
Requested VBF feature cross-check

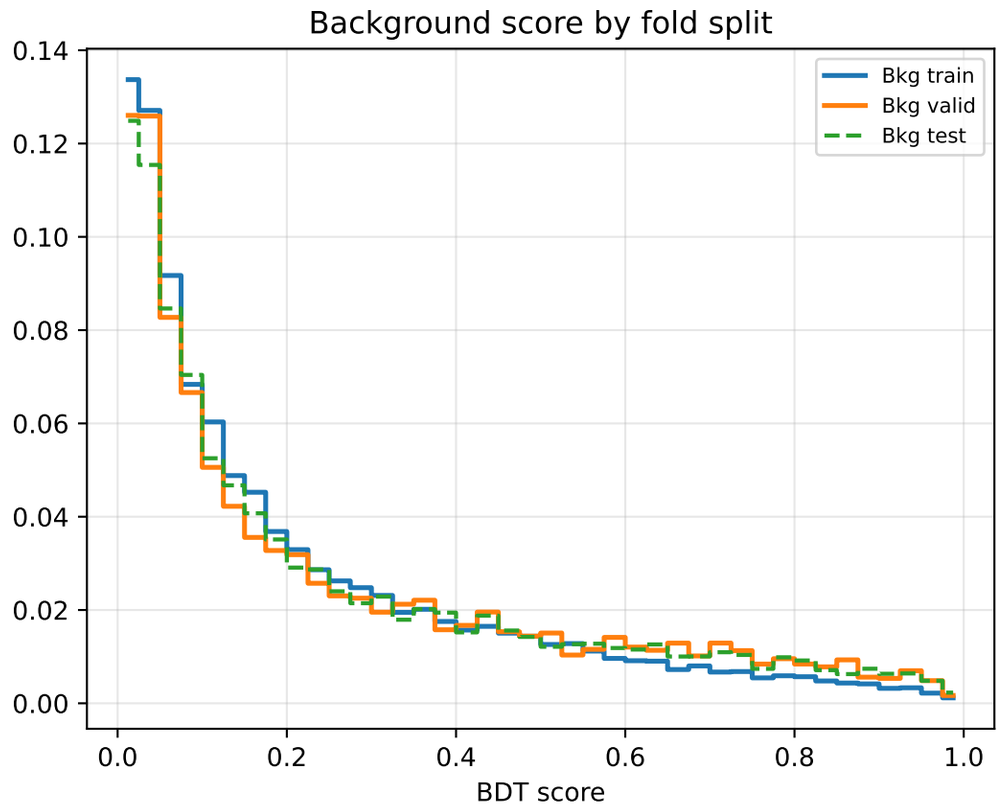
requested	status	resolved_as	notes
jet1_qgl	missing	-	not found in tree branches
jet1_eta	exact	jet1_eta	
jet2_eta	proxy	y_j2	rapidity, not eta
jet1_pt	alias	pt_j1	
jet2_pt	alias	pt_j2	
mjj	missing	-	not found in tree branches
dEta_jj	alias	deta_jj	
dPhi_jj	alias	dphi_jj	
mmj_min_dEta	proxy	deta_mumu_jj	dimuon-dijet deta, not min(mu,jet)
mmj_min_dPhi	proxy	dphi_mumu_jj	dimuon-dijet dphi, not min(mu,jet)
pt_centrality	proxy	z	zeppenfeld variable, not pt centrality
dimuon_pt	alias	pt_mumu	
dimuon_rapidity	alias	y_mumu	
dimuon_cos_theta_cs	alias	costheta_CS	
dimuon_phi_cs	alias	phi_CS	
dimuon_mass_resolution	missing	-	not found in tree branches
mu1_pt_over_mass	alias	mu1_pt_over_m	
mu2_pt_over_mass	alias	mu2_pt_over_m	
mu1_eta	exact	mu1_eta	
mu2_eta	exact	mu2_eta	
N_jets	alias	njets	
year	derived	year=2024	from --era


A) Feature importance ranking
pt_balance
z
deta_jj
njets
pt_j2
dphi_mumu_jj
deta_mumu_jj
pt_mumu
jet1_eta
pt_j1
dphi_jj
mu1_pt_over_m
mu1_eta
y_mumu
phi_CS
mu2_eta
mu2_pt_over_m
costheta_CS
source: src/drivers/9_TMVA/Outputs/Training/2024/2024_featureimportance_background.txt
0
5
10
15
20
Feature importance (gain fraction or %)


BDT score vs jet multiplicity (SR)
Njets = 2
Njets = 3
Njets >= 4
DY-like
VBF-like
Normalized entries
BDT score
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.0
0.2
0.4
0.6
0.8
1.0


5) Optimized BDT categories (high-to-low merge)
Total multibin Z_Asimov=0.461
Background
Signal
Background weighted yield
Signal weighted yield
0
2000
4000
6000
8000
10000
12000
0
1
2
3
4
5
6
7
[0.00,0.05)
[0.05,0.15)
[0.15,0.32)
[0.32,0.53)
[0.53,0.74)
[0.74,0.88)
[0.88,0.95)
[0.95,1.00)



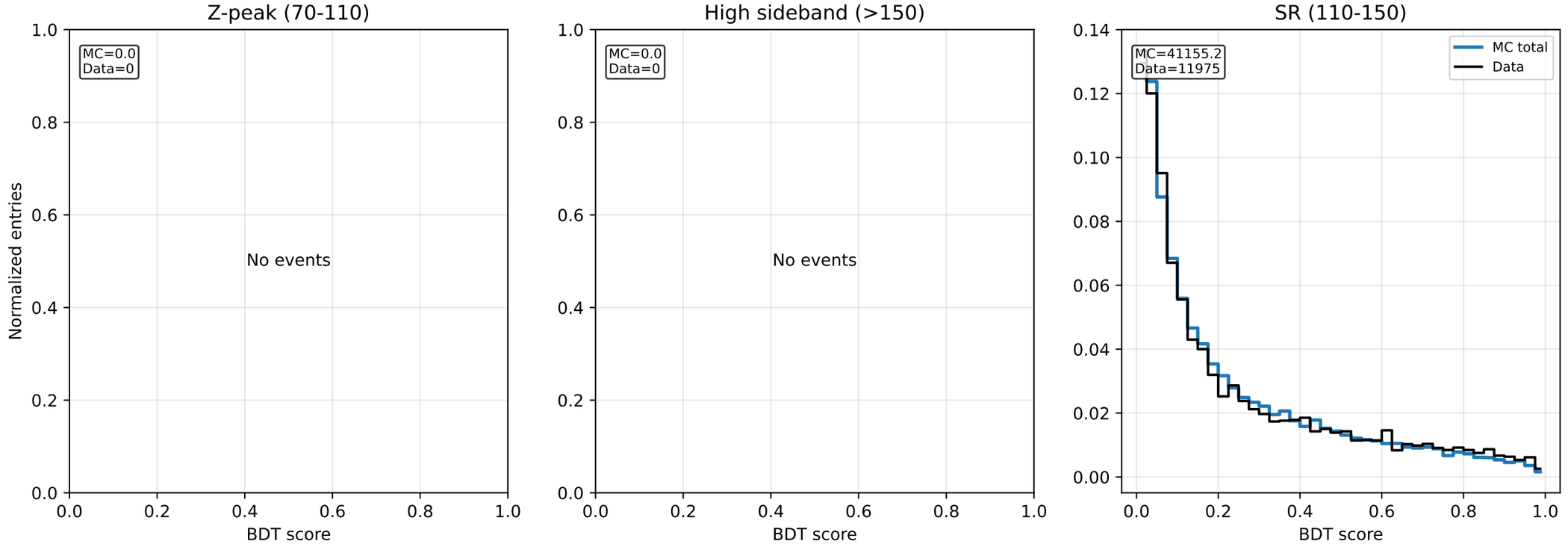


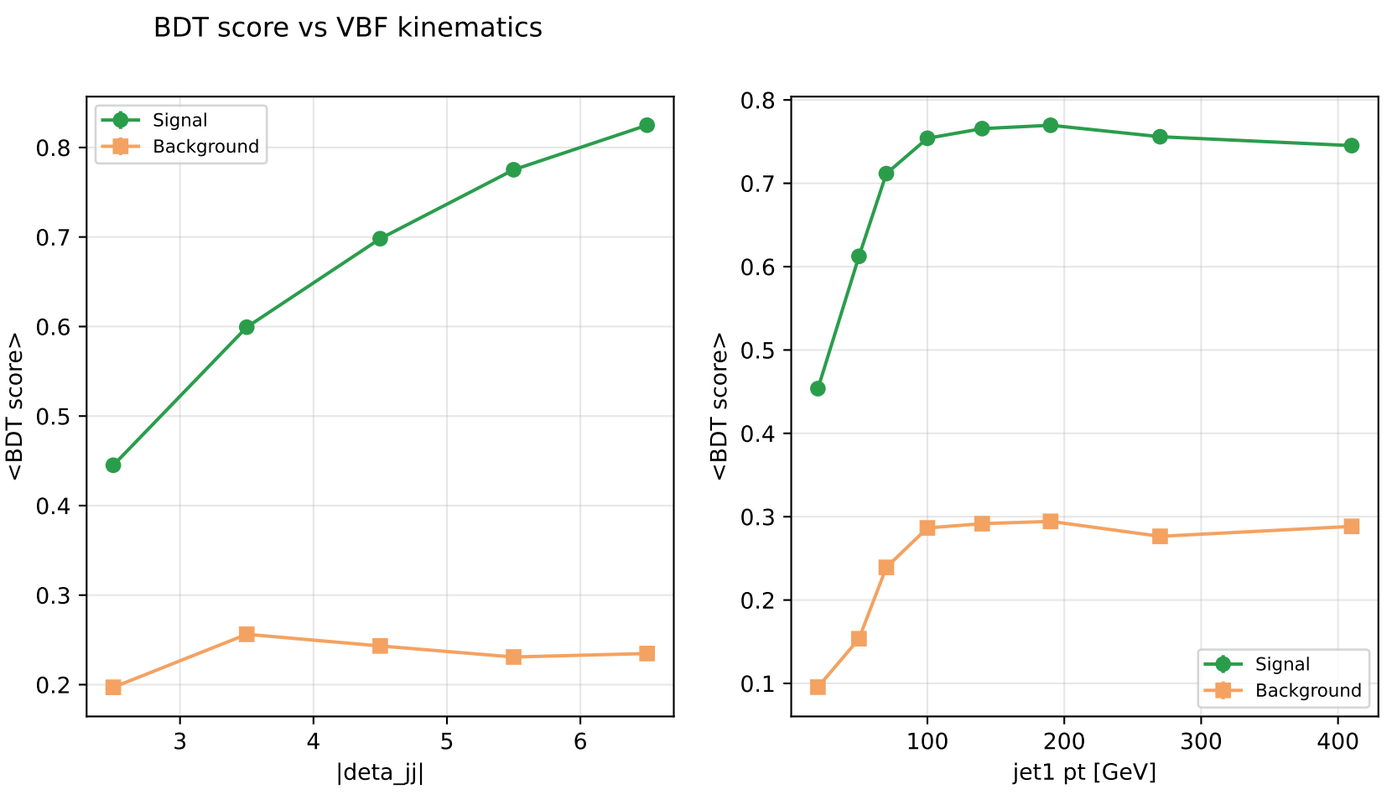
Fold-based KS checks (train vs validation)

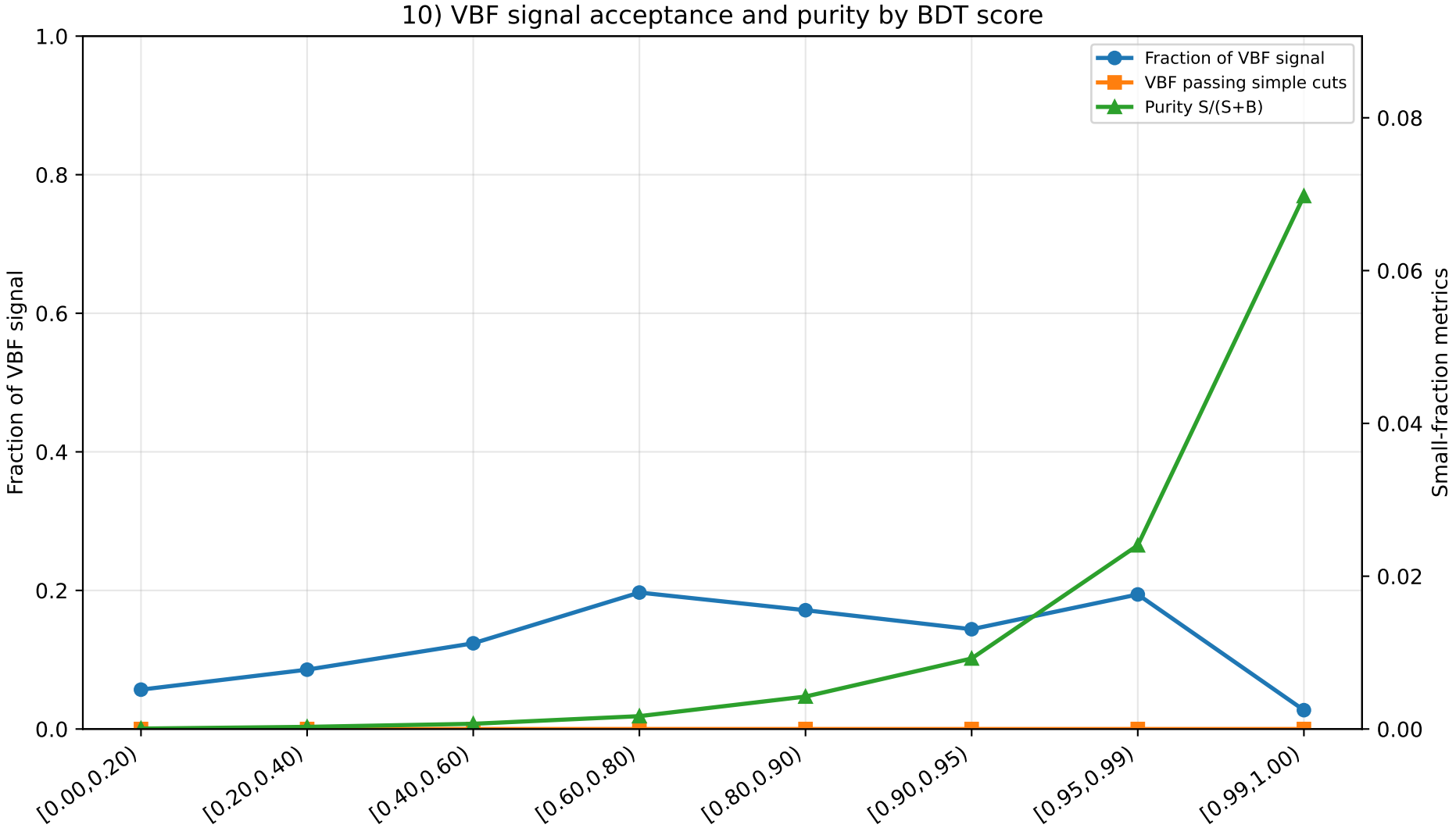
KS thresholds: Run2-loose: p>=0.01, D<=0.15; Run2-nominal: p>=0.05, D<=0.1; Run2-tight: p>=0.1, D<=0.07

Sample	N train	N valid	N test	Mean(train)	Mean(valid)	Mean(test)	KS D	KS p	Run2-loose	Run2-nominal	Run2-tight
Signal	728293	364146	364146	0.7311	0.7299	0.7310	0.0021	0.2478	**PASS**	**PASS**	**PASS**
Background	156494	78244	78240	0.2203	0.2568	0.2537	0.0127	9.933e-08	**FAIL**	**FAIL**	**FAIL**

Corr(score, njets): signal=-0.184, background=-0.045


Control-region BDT response
Z-peak (70-110)
High sideband (>150)
SR (110-150)
MC=0.0
Data=0
No events
MC=41155.2
Data=11975
MC total
Data
Normalized entries
BDT score


BDT score vs VBF kinematics
Signal
Background
<BDT score>
|deta_jj|
jet1 pt [GeV]


10) VBF signal acceptance and purity by BDT score
Fraction of VBF signal
VBF passing simple cuts
Purity S/(S+B)
Fraction of VBF signal
Small-fraction metrics
1.0
0.8
0.6
0.4
0.2
0.0
0.08
0.06
0.04
0.02
0.00
[0.00,0.20)
[0.20,0.40)
[0.40,0.60)
[0.60,0.80)
[0.80,0.90)
[0.90,0.95)
[0.95,0.99)
[0.99,1.00)

Validation Summary

Era: 2024
Inputs: data_files=1 mc_groups=7
Signal groups: SIGNAL
VBF groups: SIGNAL
SR weighted yields: S=25.337, B=41129.856
KS profiles: Run2-loose: p>=0.01, D<=0.15; Run2-nominal: p>=0.05, D<=0.1; Run2-tight: p>=0.1, D<=0.07
Feature importance source: src/drivers/9_TMVA/Outputs/Training/2024/2024_featureimportance_background.txt
Optimized categories: n=8, Z_Asimov=0.4606
KS(signal train-vs-valid): D=0.0021, p=0.2478
KS(bkg train-vs-valid): D=0.0127, p=9.933e-08
KS profile checks (Signal): Run2-loose=PASS, Run2-nominal=PASS, Run2-tight=PASS
KS profile checks (Background): Run2-loose=FAIL, Run2-nominal=FAIL, Run2-tight=FAIL
Corr(score, njets) signal=-0.184, background=-0.045
Highest BDT-bin purity=0.070